BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Work Integrated Learning Programmes Division

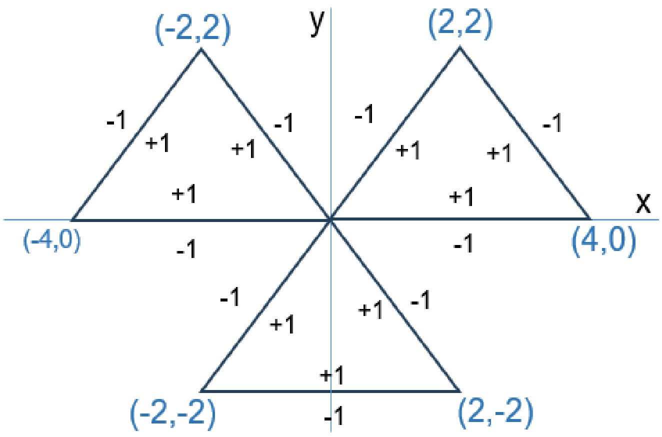
Cluster Programme - M. Tech in AI & ML

II Semester , 2023 – 24(July,2024)

Mid semester Examination (**MAKEUP**)

Course No	:	AIMLC ZC511
Course Title	:	DEEP NEURAL NETWORK
Nature of Exam.	:	Closed Book
Weightage	:	30 Marks
Duration	:	120 minutes
Date	:	28st July,2024_10 AN

Number of questions:6
Number of Pages: 2

Q. No	Question	Marks
Q.1.	Design a fully connected multilayer perceptron network with minimum number of hidden layers and hidden nodes required to classify the below decision boundary with 100% accuracy. (x, y) are input features and target classes are either +1 or -1 as shown in the figure.Step activation functions are used at all nodes, i.e., output=+1 if total weighted input >= threshold (mention inside node) at a node, else output = -1. 	**6M**
Q.2.	(a) Given an input matrix of size 8×8 and a kernel size of 3×3 with a stride of 1 and no padding, determine the output shape after applying the convolution operation. [2 M] (b) If there are 5 such filters applied, what will be the output shape? [1 M] (c) Apply a 4×4 max-pooling filter with a stride of 2 to the output from part (b). What will be the final output shape? [2 M]	**5M**

Q.3.	Mr Ram has a dataset **data_1.csv** with 1 million labelled training examples for classification, and dataset **data_2.csv** with 100 labelled training examples. Mr Rakesh trains a model from scratch on **data_2.csv**. Mr Ram decided to train on **data_1.csv**, and then apply transfer learning to train on **data_2.csv**. Differentiate between these approaches and the advantages / problems both of them faces and how to solve them, if possible? State one problem your friend is likely to find with his approach. How does your approach address this problem?	**4M**
Q.4.	Given an error function $E(w_1,w_2) = 3w_1^2 + 4w_2^2 + 2w_1w_2$,different variants of gradient descent can be used to minimize the error with respect to w_1 and w_2. Assume initial weights are $(w_1, w_2) = (0.5,0.5)$ at time $t-1$. (a) Calculate the gradients $\partial E/\partial w1$ and $\partial E/\partial w2$. [3 M] (b) Using the standard gradient descent with a learning rate $\eta=0.1$, compute the weight updates and the new weights at time t. [2 M]	**5M**
Q.5	a) Discuss any two benefits of using convolution layers instead of fully connected layers for image classification [2 M] b) Suppose that you are training a deep learning algorithm on a given data set. You observed that accuracy of the algorithm is decreasing after few epochs. Then what is your interpretation of this and how to address this situation, if possible [3 M]	**5M**
Q.6.	a) List the hyper parameters used for training a neural network. [2 M] b) We know that ANN help us in image classification. If so, why CNN is preferred over ANN for image classification [3 M]	**5M**